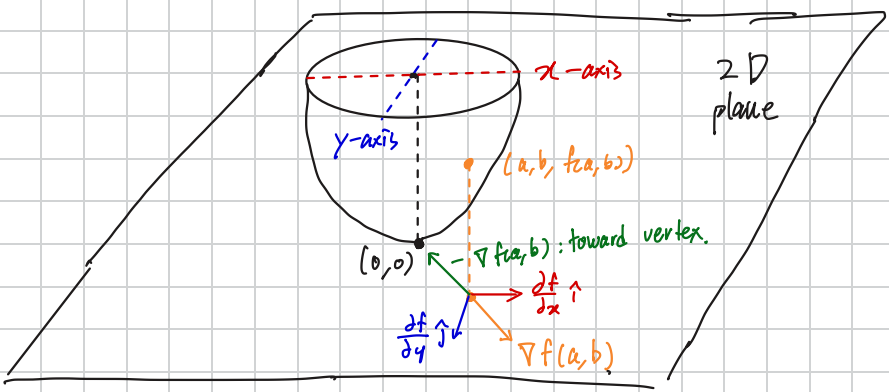
< Gradient >

$$\nabla = \frac{\partial}{\partial x}\hat{i} + \frac{\partial}{\partial y}\hat{j} + \frac{\partial}{\partial z}\hat{k}$$

$$\Rightarrow \nabla f = \frac{\partial f}{\partial x}\hat{i} + \frac{\partial f}{\partial y}\hat{j} + \frac{\partial f}{\partial z}\hat{k}$$

<< e.g., 3D paraboloid $z = f(x,y)$ >> ; 2D gradient field.



$\nabla f(a,b)$ direction → increase in $f(x,y)$.

⇒ gradient descent focuses on vertex ⇒ opposite direction of $\nabla f(x,y)$.

↓

find minimum $|\nabla f(x,y)|$

b/c $\frac{\partial f}{\partial x}$ & $\frac{\partial f}{\partial y} = 0$ at vertex.